Q. 2

06.10.2026

(*) $Y = \overline{C}(\overline{A}\,\overline{B}\,\overline{D} + D) + A\overline{B}C + \overline{D}$; Simplify them using K-map.

Solution:

Given,

$$Y = \overline{C}(\overline{A}\,\overline{B}\,\overline{D} + CD) + A\overline{B}C + \overline{D}$$

$$= \overline{A}\,\overline{B}\,\overline{C}\,\overline{D} + \overline{C}D + A\overline{B}C + \overline{D}$$

AB \ CD	$\overline{C}\,\overline{D}$ 00	$\overline{C}D$ 01	CD 11	$C\overline{D}$ 10
$\overline{A}\,\overline{B}$ 00	1	1	0	1
$\overline{A}B$ 01	1	1	0	1
AB 11	1	1	0	1
$A\overline{B}$ 10	1	1	1	1

$\therefore Y = \overline{C} + A\overline{B} + C\overline{D}$

$= A\overline{B} + C\overline{D} + \overline{C}$

$Y = \overline{C} + A\overline{B} + \overline{D}$

$= \overline{C} + \overline{D} + A\overline{B}$

(Ans)